

9647/01

20 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and shade your exam number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

This document consists of **17** printed pages and 1 blank page.

[Turn over

SECTION A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1. Which statement about one mole of a metal is always correct?
- A** It contains the same number of atoms as $\frac{1}{2}$ mol of hydrogen gas.
- B** It contains the same number of atoms as $\frac{1}{12}$ mol of ^{12}C .
- C** It has the same mass as 1 mol of hydrogen atoms.
- D** It is liberated by 1 mol of electrons.
2. An element **Y** can exist in a few oxidation states. 0.01 mol of Y^{2+} requires 0.004 mol of acidified KMnO_4 for complete reaction. The half equation for reduction of MnO_4^- is:



What is the final oxidation state of **Y**?

- A** +4 **B** +3 **C** 0 **D** -1
3. 0.0337 g of an organic compound when vapourised completely at 100 kPa and 300 K occupied a volume of 20 cm^3 . When this volume of vapour was completely burnt in excess oxygen, 40 cm^3 of carbon dioxide and 20 cm^3 of water vapour were formed. All volumes were measured at the same temperature and pressure.

What is the formula of the organic compound?

- A** C_2H_2
- B** C_3H_6
- C** CH_2CO
- D** CH_3CHO
4. Why is the first ionisation of sulfur lower than that of phosphorus?
- A** Sulfur is more electronegative than phosphorus.
- B** Sulfur forms anions more readily.
- C** The electron to be lost in sulfur is paired with another electron.
- D** The sulfur atom is larger than the phosphorus atom.

5. Use of the Data Booklet is relevant to this question.

Four elements, **W**, **X**, **Y** and **Z**, are in the fourth period with consecutive atomic numbers. The table shows the number of unpaired electrons in each atom in its ground state.

Element	W	X	Y	Z
Unpaired electrons	2	1	0	1

What is the identity of element **W**?

- A** Ti **B** Ni **C** Ge **D** Se

6. Three substances **R**, **S** and **T** have the following physical properties.

Substance	mp / °C	bp / °C	Electrical conductivity in	
			solid state	liquid state
R	801	1413	poor	good
S	2852	3600	poor	good
T	3550	4827	good	not known

What could be the identities of **R**, **S** and **T**?

	R	S	T
A	NaCl	KCl	Cu
B	NaBr	CaO	SiO ₂
C	NaCl	MgO	C(graphite)
D	NaBr	CaO	C(diamond)

7. Which pair of compounds below satisfies the following conditions?
- (i) The first compound has a larger bond angle than the second compound.
- (ii) The first compound is more polar than the second compound.

	first compound	second compound
A	SO ₂	HCN
B	CO ₂	BeCl ₂
C	IF ₃	PH ₃
D	NF ₃	SF ₆

8. A student used a simple calorimeter to determine the enthalpy change of combustion of propan-1-ol. The following results were obtained:

mass of propan-1-ol burnt	0.60 g
mass of water heated	200 g
initial temperature of water	21.0 °C

Given that the enthalpy change of combustion of propan-1-ol is $-2021 \text{ kJ mol}^{-1}$, and that the heating is only 80% efficient, what would be the final temperature of the water?

Assume that the specific heat capacity of water is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

[M_r of propan-1-ol = 60]

- A** 27.4 °C **B** 29.0 °C **C** 40.2 °C **D** 45.1 °C
9. The table below gives the standard enthalpy change of hydrogenation, ΔH , of three compounds to form cyclohexane:

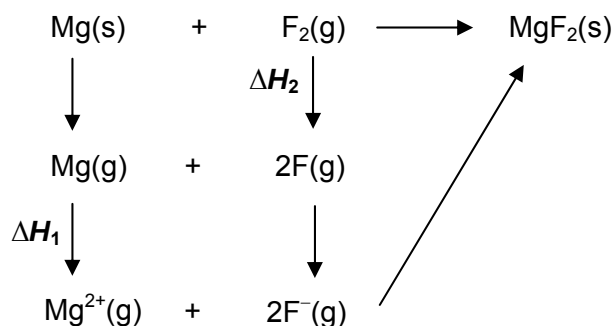
Compound	$\Delta H / \text{kJ mol}^{-1}$
benzene	-206
cyclohexene	-120
cyclohexatriene	-360

Which of the following statements is correct?

- A** Cyclohexatriene is more stable than benzene.
B Benzene has lower energy content than cyclohexatriene.
C The C=C bond energy in cyclohexene is weaker than that in cyclohexatriene.
D The enthalpy change of atomisation of benzene is smaller than that of cyclohexatriene.
10. Each of the following equilibria is subjected to two changes carried out separately:
 (i) the pressure is reduced at constant temperature;
 (ii) the temperature is increased at constant pressure.
 For which equilibrium will both of these changes result in an increase in the proportion of products?

- A** $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) = 2\text{HI}(\text{g})$ $\Delta H = +53 \text{ kJ mol}^{-1}$
B $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) = 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ $\Delta H = -950 \text{ kJ mol}^{-1}$
C $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) = 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$
D $\text{N}_2\text{O}_4(\text{g}) = 2\text{NO}_2(\text{g})$ $\Delta H = +57 \text{ kJ mol}^{-1}$

11. The diagram shows the Born-Haber cycle for the formation of magnesium fluoride.



Which of the following shows the correct energy changes?

	ΔH_1	ΔH_2
A	First and second electron affinity	$2 \times$ Enthalpy change of atomisation
B	First and second ionisation energy	$\frac{1}{2} \times$ Bond energy
C	First and second electron affinity	$\frac{1}{2} \times$ Bond energy
D	First and second ionisation energy	$2 \times$ Enthalpy change of atomisation

12. A current is passed through two cells connected in series. The first cell contains $\text{XCl}_2(\text{aq})$ while the second cell contains $\text{YCl}(\text{aq})$. The mass of **X** and **Y** deposited at the respective electrodes is in the ratio of 1:3.

What is the ratio of the relative atomic mass of **X** : relative atomic mass of **Y**?

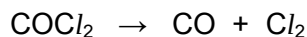
- A** 1 : 6 **B** 2 : 3 **C** 3 : 2 **D** 6 : 1

13. Which of the following is the least soluble compound?

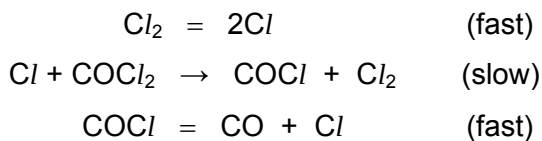
Compound	Solubility product, K_{sp}
AgBr	5.35×10^{-13}
AgI	1.50×10^{-16}
$\text{Cu}(\text{OH})_2$	2.60×10^{-19}
$\text{Pb}(\text{OH})_2$	1.43×10^{-20}

- A** AgBr **B** AgI **C** $\text{Cu}(\text{OH})_2$ **D** $\text{Pb}(\text{OH})_2$

14. Phosgene gas, COCl_2 , is a toxic gas which was extensively used in combat during World War I. The mechanism for the decomposition of COCl_2



is thought to be as follows:



Based on this mechanism, what is the rate equation for this reaction?

- A Rate = $k[\text{COCl}_2][\text{Cl}]$
 B Rate = $k[\text{COCl}_2][\text{Cl}_2]^2$
 C Rate = $k[\text{COCl}_2][\text{Cl}_2]^{\frac{1}{2}}$
 D Rate = $k[\text{COCl}_2][\text{Cl}_2]$
15. Three examples of industrial reactions in which sulfuric acid is used are shown below.
- Reaction 1: $\text{Al}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
 Reaction 2: $\text{Cu} + \text{H}_2\text{SO}_4 \rightarrow \text{CuO} + \text{SO}_2 + \text{H}_2\text{O}$
 Reaction 3: $(\text{CH}_3)_3\text{COH} + \text{H}_2\text{SO}_4 \rightarrow (\text{CH}_3)_2\text{C}=\text{CH}_2 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$

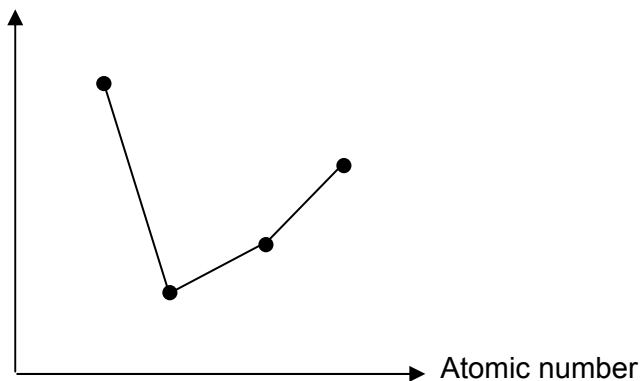
What is the function of sulfuric acid in each reaction?

	Reaction 1	Reaction 2	Reaction 3
A	Acidic	Oxidising	Dehydrating
B	Acidic	Acidic	Dehydrating
C	Dehydrating	Oxidising	Catalytic
D	Dehydrating	Acidic	Catalytic

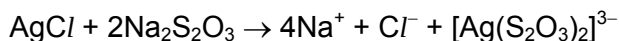
16. Element **D** has a high melting point of 1414 °C. Its oxide does not dissolve in dilute sodium hydroxide solution.
- What is likely to be element **D**?

- A Silicon
 B Sodium
 C Aluminium
 D Phosphorus

17. Which property when plotted against atomic number gives a graph of the following shape?

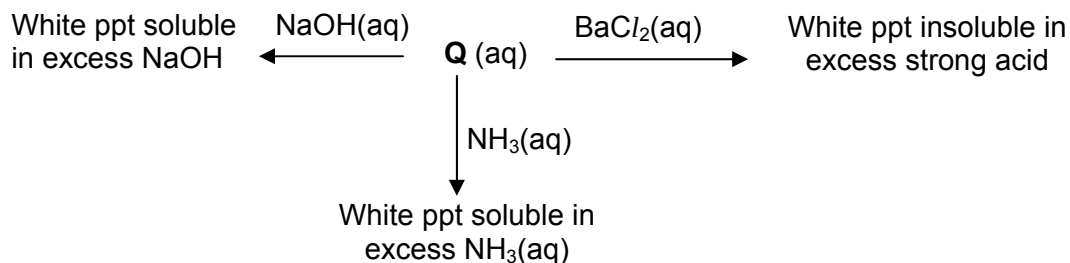


- A Radius of F^- , Ne, Na^+ , Mg^{2+}
 B Boiling point of HF, HCl, HBr, HI
 C Second ionisation energy of Ne, Na, Mg, Al
 D pH of the aqueous solution formed by Na_2O , MgO, Al_2O_3 , SiO_2
18. In black and white photographic film, light converts silver chloride into metallic silver. After the film has been developed, the unreacted silver chloride is removed by using a fixer such as sodium thiosulfate. The reaction of silver chloride with sodium thiosulfate is shown below.



Which process does **not** occur in the above reaction?

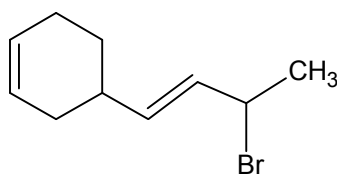
- A Dissolving of silver ions
 B Formation of dative covalent bonds
 C Formation of complex
 D Oxidation of silver ions
19. The diagram shows some reactions of salt **Q**.



What is the identity of **Q**?

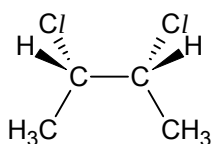
- A $Al_2(SO_4)_3$ B $Pb(NO_3)_2$ C $ZnSO_3$ D $ZnSO_4$

20. How many stereoisomers does compound **G** have?



compound **G**

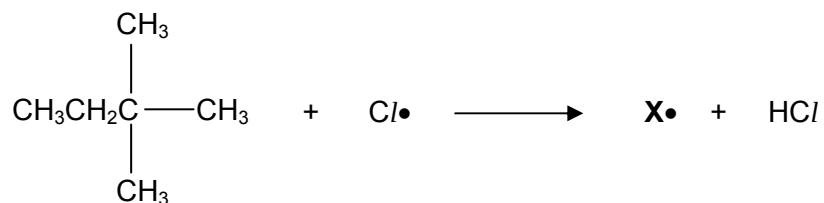
- A** 3 **B** 4 **C** 8 **D** 16
21. The structure below represents *meso*-2,3-dichlorobutane.



Which of the following statements explains why this isomer of 2,3-dichlorobutane does not show optical activity?

- A** The molecule is a racemic mixture.
B The molecule contains a plane of symmetry.
C The molecule does not contain any chiral carbon.
D The molecule has net dipole moment which cancels out the rotation of plane-polarised light.
22. When heated with chlorine, the hydrocarbon 2,2-dimethylbutane undergoes free radical substitution.

In a propagation step, the free radical **X•** is formed.



How many different forms of **X•** are possible?

- A** 2 **B** 3 **C** 4 **D** 5

23. Calcium ethanoate produces propanone when heated as shown in the equation below.

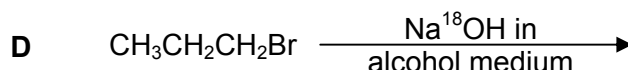
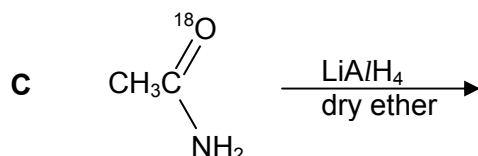
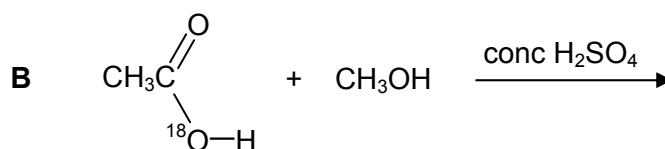
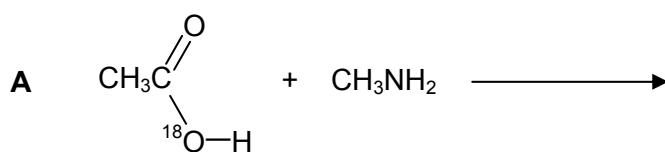


When this reaction is carried out by heating a 1:1 molar mixture of calcium propanoate, $(\text{CH}_3\text{CH}_2\text{CO}_2)_2\text{Ca}$, and calcium methanoate, $(\text{HCO}_2)_2\text{Ca}$, a mixture of three carbonyl compounds is produced.

Which of the following carbonyl compounds is **not** formed?

- | | |
|--|---|
| A HCHO | B $\text{CH}_3\text{CH}_2\text{CHO}$ |
| C $\text{CH}_3\text{CH}_2\text{COCH}_3$ | D $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ |

24. Which of the following reactions produces an organic product containing ^{18}O atom?

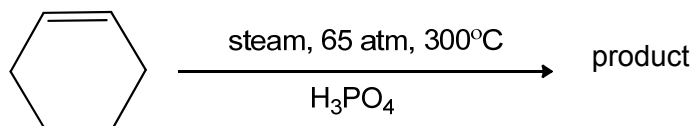


25. **J** is insoluble in water but dissolves readily in cold hydrochloric acid. Evaporation of this solution yields a crystalline solid.

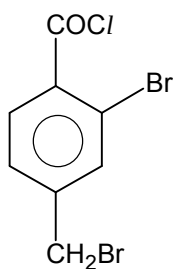
Which of the following could be **J**?

- | | | | |
|-----------------------------------|--|--|---|
| A CH_3NH_2 | B $\text{CH}_3\text{CO}_2\text{CH}_3$ | C $\text{C}_6\text{H}_5\text{CN}$ | D $\text{C}_6\text{H}_5\text{CH}_2\text{NHCH}_3$ |
|-----------------------------------|--|--|---|

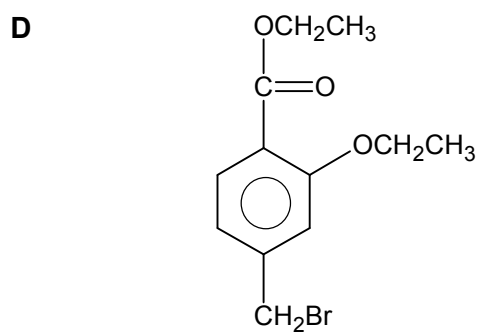
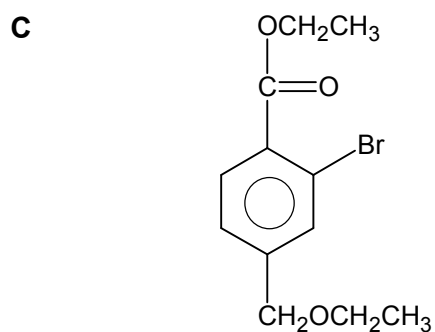
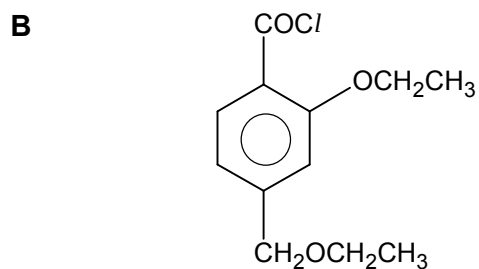
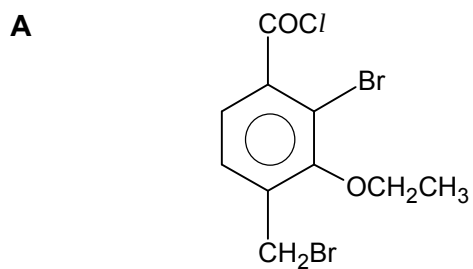
26. Which reagent can be used to indicate that the following reaction has taken place?



- A $\text{K}_2\text{Cr}_2\text{O}_7$ B Br_2 C KMnO_4 D neutral FeCl_3
27. What is a likely product when compound **K** reacts with excess sodium ethoxide ($\text{CH}_3\text{CH}_2\text{ONa}$)?

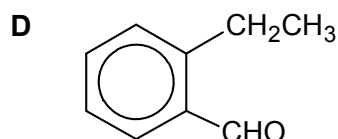
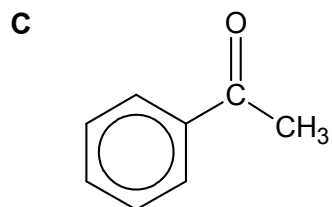
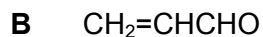
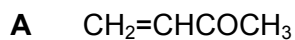


compound **K**

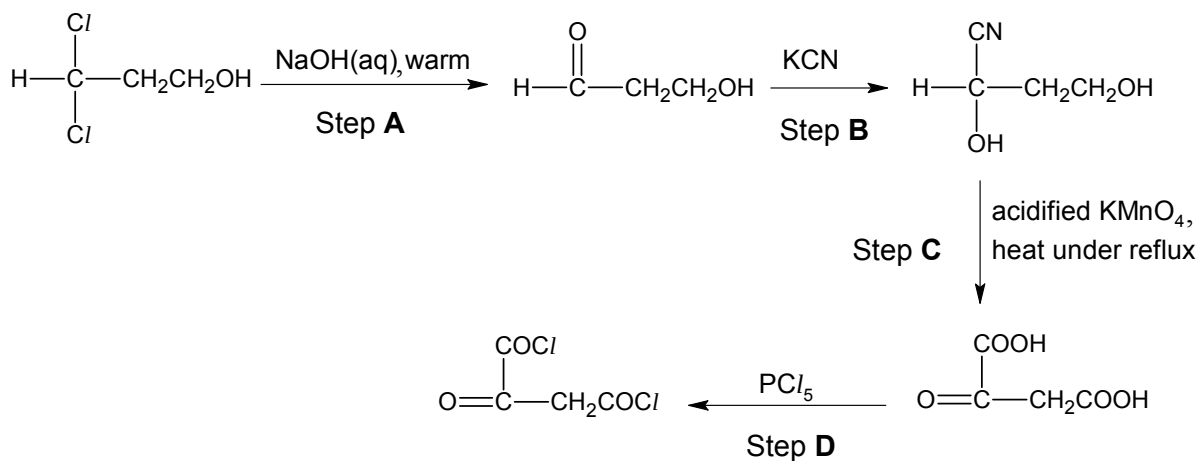


28. Smoke from a bonfire contains a compound that causes irritation to the eyes. This compound produces CO_2 when oxidised by KMnO_4 and produces a red brown precipitate when bubbled into Fehling's solution.

What is a possible structure of the compound?

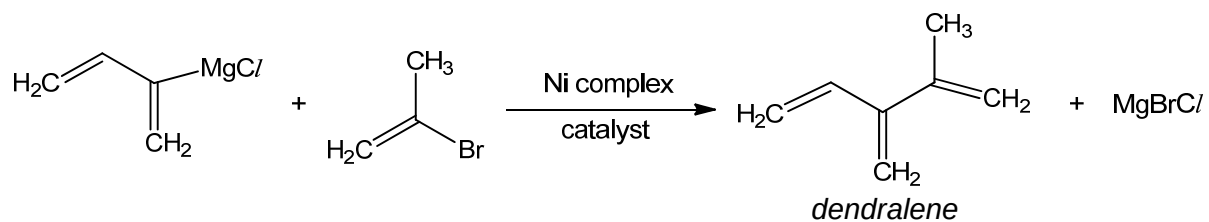


29. A reaction scheme is given below.

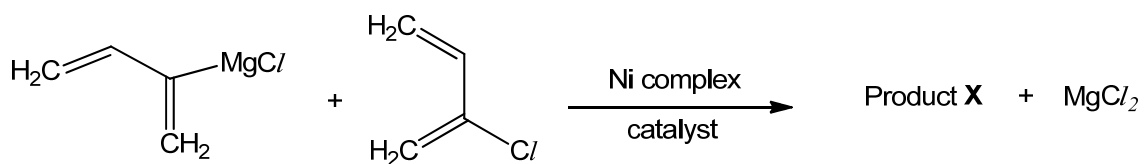


Identify the step in which a wrong reagent was used.

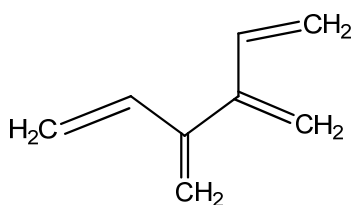
30. The breakthrough in nickel chemistry was the discovery of a nickel complex that catalysed the reaction that forms *dendralene*.



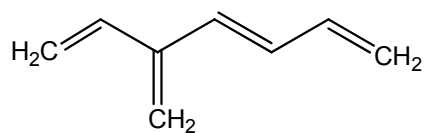
What is the structural formula of product **X** in the following reaction?



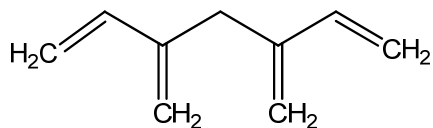
A



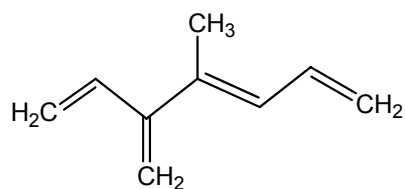
B



C



D



SECTION B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct.

The responses **A** to **D** should be selected on the basis of

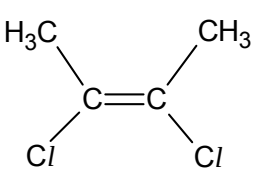
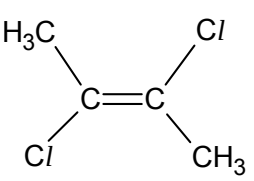
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31.** A group of students attempted to estimate the concentration of a solution of Fe^{2+} by pipetting fixed volumes of the solution into a flask, adding an excess of dilute sulfuric acid, and then titrating with a standard solution of potassium manganate(VII) from a burette. The volume of KMnO_4 solution required by one student was 0.20 cm^3 higher than that of the other students.

Which of the following are possible explanations for this discrepancy?

- 1** The titration flask was rinsed with the solution of Fe^{2+} instead of water before titration.
 - 2** The last drop of Fe^{2+} solution was blown from the pipette into the flask.
 - 3** The burette was rinsed with water instead of the solution of KMnO_4 before titration.
- 32.** In which of the following pairs, does the first compound have a higher boiling point than the second compound?

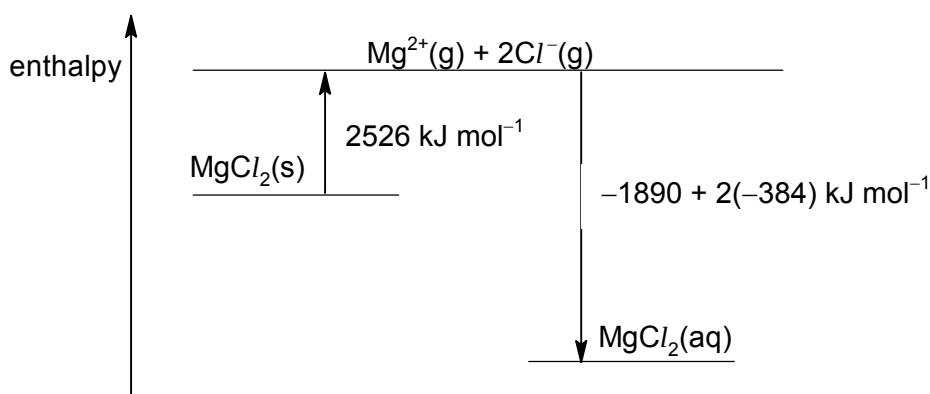
	first compound	second compound
1	$\text{H}_2\text{NCH}_2\text{COOH}$	$\text{H}_2\text{NCH}_2\text{CH}_2\text{OH}$
2		
3	$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	$(\text{CH}_3)_3\text{N}$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

33. The following energy level diagram represents the dissolving of anhydrous magnesium chloride in a large volume of water.



Which statements about the process are correct?

- 1 The lattice energy of magnesium chloride is $-2526 \text{ kJ mol}^{-1}$.
 - 2 The enthalpy change of hydration of the chloride ion is -384 kJ mol^{-1} .
 - 3 The enthalpy change of solution of anhydrous magnesium chloride is -132 kJ mol^{-1} .
34. Pain is often felt when a piece of aluminium foil touches a dental amalgam filling in a tooth because an electric current momentarily flows. The amalgam contains tin. The standard electrode potentials are as follows:



Which of the following are the features of the cell obtained?

- 1 The aluminium foil acts as the negative electrode.
- 2 The e.m.f of the cell is $+1.79 \text{ V}$.
- 3 Sn^{2+} ions are momentarily released into the saliva in the mouth.

The responses **A** to **D** should be selected on the basis of

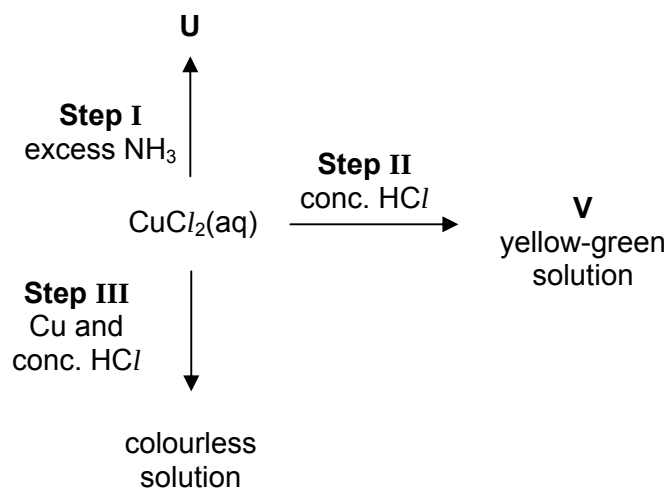
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

35. Which of the following can take place when a hot solution of brine is electrolysed using carbon electrodes in the absence of a diaphragm?

- 1 The main product is sodium chlorate(V).
- 2 Oxygen gas is evolved at the anode.
- 3 The pH of the electrolyte decreases.

36.



Which of the following statements are correct?

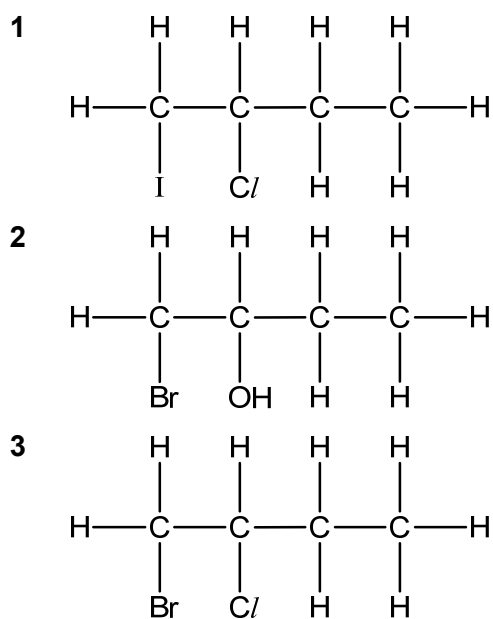
- 1 **U** is likely to be an octahedral complex with the formula $[\text{Cu}(\text{NH}_3)_2(\text{H}_2\text{O})_4]^{2+}$.
- 2 **V** is likely to be CuCl_4^{2-} .
- 3 CuCl_2 undergoes reduction in **Step III**.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 37.** Which products may be formed when but-1-ene reacts with a mixture of IBr and aqueous NaCl?



- 38.** Which of the following statements about the reaction between carbonyl compounds and hydrogen cyanide are correct?

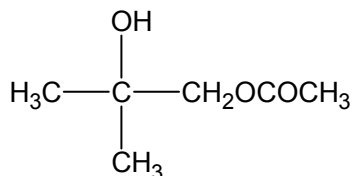
- 1** The slow step involves the formation of a C–C bond.
- 2** HCN acts as a Bronsted acid.
- 3** The reaction is catalysed by a small amount of H^+ ion.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

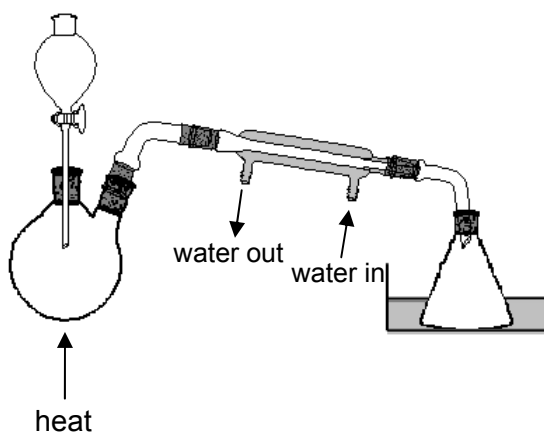
No other combination of statements is used as a correct response.

39. Which reagent reacts with the compound shown below?



- 1** dilute sulfuric acid
- 2** alkaline aqueous iodine
- 3** 2,4-dinitrophenylhydrazine

40. The diagram shows some laboratory apparatus.



Which preparations could this apparatus be used for?

- 1** 1,2-dibromoethane from ethene and bromine
- 2** ethanal, from ethanol, sodium dichromate(VI) and sulfuric acid
- 3** bromoethane, from ethanol, sodium bromide and concentrated sulfuric acid

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